1 **(a)** Fig. 1.1. is a speed–time graph for the first 5 minutes of a bus journey.

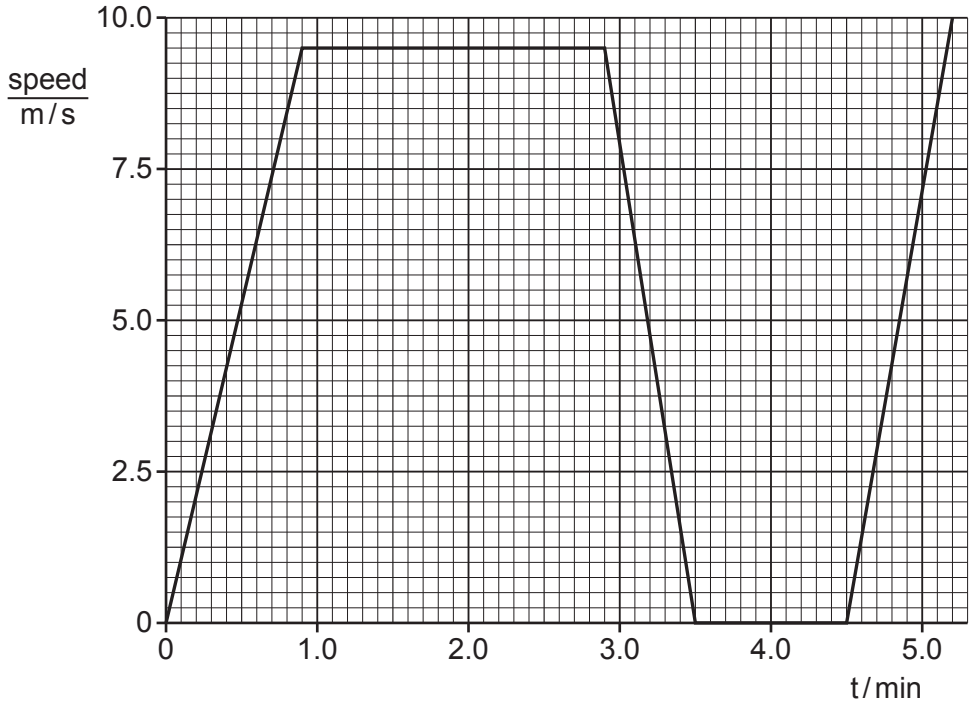


Fig. 1.1

Describe the motion between:

1. t = 0.90 min and t = 2.9 min .....

2. t = 2.9 min and t = 3.5 min .....

3. t = 3.5 min and t = 4.5 min .....

[3]

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(b) Another bus travels at a speed of 8.9 m / s. The brakes apply a constant force and the bus stops in a distance of 23 m. This bus has a mass of 18 000 kg.

(i) Calculate the kinetic energy of the bus before the brakes are applied.

kinetic energy = ... [2]

(ii) Calculate the force applied to stop the bus.

force = ... [3]

[Total: 8]

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